Highlights

Mostly ranking, partly threshold: decomposing and repairing the cross-corpus generalisation gap of misinformation detectors

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- Ten audited corpora and seven detectors, trained on one corpus and tested on all.
- About 70% of the cross-corpus F1 gap is lost ranking; the rest is the threshold.
- EM label-shift correction worsens the threshold because calibration breaks.
- Flagging the source base rate fixes most of the threshold loss without labels.
- Bengali clickbait transfers across publishers; English veracity does not.

Mostly ranking, partly threshold: decomposing and repairing the cross-corpus generalisation gap of misinformation detectors

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Abstract

Fake-news detectors score near-perfectly on their training corpus and much worse on any other. Cross-corpus studies report this with accuracy or F1, which mixes two failures: ranking new items badly, and ranking them well but thresholding them wrongly. We separate the two. Ten public corpora in English and Bengali (articles, headlines, claims, tweets) are audited for label direction and near-duplicates, and seven detectors, including a partially fine-tuned RoBERTa, are trained on one corpus and tested on all, with repeated seeds, bootstrap intervals and permutation controls. The audit shows that the two most used article corpora code their labels in opposite directions, which turns moderate transfer into an apparent below-chance collapse. Mean ROC-AUC falls from 0.914 in-domain to 0.567 across corpora. Of the macro-F1 that is lost, 70% (95% corpus-bootstrap interval [63%, 79%]) is a ranking deficit and the rest a misplaced operating point. EM prior adaptation, the usual label-shift correction, makes the operating point worse, because calibration does not survive the change of corpus. Fixing the share of items flagged at the source base rate recovers 81% of the operating-point loss without target labels; we show when this rule is exact and that it degrades once the base rate itself shifts. Pooling six training corpora is the only representation change that improves held-out ranking; removing corpus identity lowers it, because corpus and label are entangled. Bengali clickbait, by contrast, transfers across publishers and partly across languages. Code, per-item scores and a reporting protocol are released.

Keywords: misinformation detection, fake news, cross-corpus generalisation, dataset audit, label shift, shortcut learning, Bengali

1. Introduction

False stories spread farther and faster than true ones on social platforms (Vosoughi et al., 2018), and public health, elections and trust in institutions pay for that difference (Lazer et al., 2018). Automatic detection is one response. On the standard benchmarks it looks close to solved, with content-based classifiers

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routinely reporting accuracies above 95% (Shu et al., 2017; Zhou and Zafarani, 2020; Kaliyar et al., 2021). But a detector is always deployed on data it was not trained on, and a growing body of work shows that these scores do not survive a change of corpus (Pérez-Rosas et al., 2018; Janicka et al., 2019; Suprem and Pu, 2022; Dell'Oglio et al., 2026), partly because some of them reflect dataset artefacts rather than veracity (Asr and Taboada, 2019; Thibault et al., 2025; Verma, 2026).

What is still missing is an account of *why* the gap arises that also tells a practitioner what to do about it. A detector can fail on a new corpus in two ways. It can *order* the new items badly, giving fabricated stories no higher scores than genuine ones. Or it can order them well and still put its decision threshold in the wrong place, because the new corpus has a different base rate or because its scores come out on a different scale. The first failure calls for a better representation. The second only calls for a better operating point, which is far cheaper. Accuracy and F1, the metrics nearly every cross-domain study reports, cannot tell the two apart.

This paper separates them. We assemble ten public corpora in two languages, covering four text units (articles, headlines, claims and tweets) and three constructs (veracity, rumour and clickbait). We audit them for label direction and for duplication within and across corpora, then train seven detectors on each corpus and test them on every other, with repeated seeds, bootstrap confidence intervals and permutation controls. The macro-F1 a detector loses on a new corpus is split into a ranking deficit, which no threshold can recover, and an operating-point loss, which the right threshold would recover. Each part is then attacked with a matching label-free repair. The contributions are as follows.

1. **An audited multi-corpus benchmark.** The two most used article corpora code their labels in opposite directions (ISOT: 1 = fake; WELFake: 1 = real). Simple signature probes reveal the inversion, which otherwise looks like a below-chance collapse of transfer; it did in our own first pipeline. Near-duplicate detection finds label-conflicting headlines in GossipCop and source tweets shared between PHEME and Twitter15/16 (Section 3).
2. **A decomposition of the transfer gap.** Across 42 directed corpus pairs, 70% of the lost macro-F1 is a ranking deficit (corpus-bootstrap interval [63%, 79%]) and 30% is operating-point loss. Ranking remains the larger part with balanced accuracy, with a cross-fitted oracle and for every detector, including a partially fine-tuned transformer (Sections 5.2 and 5.3).
3. **A label-free operating point, and why the standard one fails.** Expectation–maximisation (EM) prior adaptation, the textbook correction for label shift, enlarges the operating-point loss (−64% recovered), because the source model's calibration breaks on the new corpus. Fixing the *rate* at which items are flagged, at the source base rate, recovers 81%. We state the conditions under which this rule is exact, show that our corpora approximately meet them, and map how it degrades when the base rate moves (Sections 4.5 and 5.4).

4. **Evidence on what the detectors learn.** A gradient-boosting model of 22 surface statistics and a publisher-identity baseline each recover much of the in-domain signal, and 64 examples suffice on the article and tweet corpora. The heaviest cues are outlet signatures and time expressions. On the ranking side, pooling six training corpora raises held-out AUC from 0.567 to 0.660, whereas making the representation invariant to corpus identity lowers it, because corpus and label are entangled in these benchmarks (Sections 5.6 and 5.7).
5. **A low-resource extension.** Bengali clickbait detection replicates the published baseline, holds up on unseen publishers and partly transfers across languages through translation, which points to a difference between cues that *are* the construct and cues that merely correlate with it (Section 5.8).
6. **A reporting protocol and a full release** of code, seeds, per-cell results and per-item scores (Section 6).

2. Related work

Cross-domain failure of fake-news detectors. Content-based detectors have been known to degrade across domains since the first multi-domain corpora appeared. Models trained on one news domain lose much of their accuracy on another (Pérez-Rosas et al., 2018; Janicka et al., 2019), and fine-tuning on one COVID-19 corpus often hurts performance on related ones (Suprem and Pu, 2022). Bozarth and Budak (2020) showed that which model looks best depends on the dataset, the evaluation type and the metric. Dell'Oglio et al. (2026) recently benchmarked twelve approaches on ten English corpora and found cross-dataset median F1 close to 0.5 for classical and fine-tuned models, with zero-shot large language models more stable; Hoy and Koulouri (2025) found stylistic and "social-monetisation" features to transfer better than token representations. These studies show *that* detectors fail to transfer, almost always through a single threshold-dependent score such as accuracy or F1. They do not ask whether the loss is one of discrimination or of a mis-set threshold, and that question is the basis of this paper.

Multi-domain and debiased detectors. A separate line of work builds detectors that cope with several domains or with drift by design. EANN trains an event discriminator adversarially so that features carry less event-specific information (Wang et al., 2018); MDFEND mixes domain experts with a domain gate (Nan et al., 2021); and ENDEF removes the contribution of an entity-only model at test time to generalise to future news (Zhu et al., 2022). All of them assume that the domain-specific part of the signal is a nuisance that can be separated from the veracity part. Our repair experiments test that assumption directly on public benchmarks and find that, in these corpora, the two are largely the same thing (Section 5.7).

Shortcuts and dataset artefacts. Shortcut learning, in which a model exploits a cue that predicts the label only in the training distribution, affects deep and shallow models alike (Geirhos et al., 2020; Gururangan

et al., 2018; McCoy et al., 2019). For misinformation, Horne and Adali (2017) and Schuster et al. (2020) showed how much of the signal is style rather than veracity, and Asr and Taboada (2019) documented topical imbalance across fake-news corpora. Verma (2026) audited ISOT in depth: subject metadata alone separate the classes perfectly, a newswire tag marks 99.2% of real articles, duplicates contaminate 19.4% of a naive test split, deleting the 1,000 highest-weight unigrams leaves F1 at 0.926, and ISOT-trained models rank LIAR claims at close to chance. Thibault et al. (2025) surveyed 75 misinformation datasets and found keyword–label correlations and temporal leakage in several, including Twitter15/16, ISOT and BanFakeNews. We build on these audits but with a different scope and aim. We audit ten corpora jointly, including duplication across corpora and label direction, and use the audit to explain and repair the transfer gap rather than only to document it.

Rumour benchmarks. Rumour-detection benchmarks have the same problem. On Twitter15/16 and PHEME, event-level spurious correlations inflate scores (Wu and Hooi, 2022), chronological splits lower them (Mu et al., 2023), and simple text baselines rival propagation-based models (Pelrine et al., 2021). We therefore treat the source posts of these corpora as one more text unit and do not re-evaluate propagation models, which these works already cover.

Label shift, quantification and invariant representations. Two families of methods from domain adaptation map onto the two parts of our decomposition. Label-shift adaptation re-estimates the target class prior from unlabelled predictions and adjusts the posteriors (Saerens et al., 2002; Lipton et al., 2018). With calibrated scores, this EM procedure is hard to beat (Alexandari et al., 2020). The quantification literature studies the closely related task of estimating class prevalence in a new sample; adjusted classify-and-count, which corrects the share of predicted positives with the classifier's source error rates, is its standard baseline (Forman, 2008). Both families assume that the class-conditional distribution of inputs, or at least of scores, stays the same between source and target. Representation methods try instead to remove domain information: correlation alignment matches feature covariances (Sun et al., 2016), adversarial training removes domain-predictive directions (Ganin et al., 2016), and iterative nullspace projection deletes linearly decodable attributes (Ravfogel et al., 2020). Domain-adaptation theory bounds target error by source error plus a divergence that a domain classifier can estimate (Ben-David et al., 2010). We know of no misinformation study that measures how much of the cross-corpus gap is a prior or threshold effect and how much a representation effect, or that tests each family on the part it is meant to fix.

Large language models. Zero-shot and few-shot large language models are less exposed to dataset-specific shortcuts because they are not trained on the benchmark (Pelrine et al., 2023; Thibault et al., 2025; Dell'Oglio et al., 2026), and they can advise smaller detectors (Hu et al., 2024). They also change the target: machine-written misinformation can be harder for both people and detectors to spot than human-

Table 1: Comparison with the closest prior audits: S&P = Suprem and Pu (2022); T = Thibault et al. (2025); V = Verma (2026); D = Dell'Oglio et al. (2026) (✓ reported, – not reported; units: a = article, h = headline, c = claim, t = tweet).

	S&P	T	V	D	**This work**
Corpora (languages)	9 (en)	75 (survey)	2 (en)	10 (en)	10 (en, bn)
Text units	mixed	mixed	a, c	mixed	a, h, c, t
Full train×test transfer matrix	✓	–	–	✓	✓
Within- and cross-corpus near-duplicate control	–	partial	✓	–	✓
Label-direction verification	–	–	–	–	✓
Threshold-free metrics (AUC, AP)	–	–	✓	–	✓
Ranking vs. operating-point decomposition	–	–	partial	–	✓
Seeds, bootstrap CIs, permutation control	–	–	✓	tests only	✓
Label-free repair evaluated	reject opt.	–	–	MERMAID	EM, rate, INLP, CORAL
Low-resource language	–	–	–	–	Bengali

written misinformation (Chen and Shu, 2024), which makes generalisation beyond the training distribution more pressing. Supervised detectors nonetheless remain the cheapest and most widely deployed option, and our decomposition applies to any detector that outputs a score, a prompted language model included.

Position of this work. Table 1 compares our study with the closest prior audits. What we add is the combination of a two-language, multi-unit transfer matrix; a decomposition that assigns the gap to ranking or to the operating point; label-free repairs aimed at each part, with conditions under which they hold; and audit findings (label inversion, cross-corpus duplication) that change measured transfer.

3. Corpora and data audit

3.1. Corpora

We use ten public corpora (Table 2). Seven share the veracity construct and form the main transfer matrix: two article corpora, ISOT (Ahmed et al., 2017, 2018) and WELFake (Verma et al., 2021a); two headline corpora from FakeNewsNet, PolitiFact and GossipCop (Shu et al., 2020), for which the public release contains titles and URLs; the claim corpus LIAR (Wang, 2017), obtained through LIAR-PLUS (Alhindi et al., 2018), binarised as {pants-fire, false} versus {mostly-true, true} with the two middle grades dropped; and the source tweets of Twitter15 and Twitter16 (Ma et al., 2017), using their false and true classes. Three further corpora extend the analysis to other constructs and to Bengali: the source tweets of PHEME rumour/non-rumour threads (Zubiaga et al., 2016, 2017); an English clickbait headline corpus (Chakraborty

Table 2: Corpora after harmonisation and deduplication. "Exact" and "near" count removed duplicates; "conflict" counts items removed because duplicates disagreed on the label; "pos." is the share labelled fake, false, rumour or clickbait. For LIAR and Twitter15/16, "raw" counts the items in the binary view (LIAR: pants-fire/false vs. mostly-true/true; Twitter: false vs. true).

Corpus	Unit	Construct	Lang.	Raw	Exact	Near	Conflict	Final	Pos.
ISOT	article	veracity	en	38,833	0	79	0	38,749	0.46
WELFake	article	veracity	en	24,530	0	815	6	23,708	0.43
PolitiFact	headline	veracity	en	1,056	71	3	6	976	0.43
GossipCop	headline	veracity	en	22,140	1,174	26	380	20,560	0.23
LIAR	claim	veracity	en	8,090	21	11	3	8,055	0.44
Twitter15	tweet	veracity	en	742	55	14	0	673	0.51
Twitter16	tweet	veracity	en	412	51	3	0	358	0.46
PHEME	tweet	rumour	en	5,802	124	98	25	5,555	0.34
ClickbaitEN	headline	clickbait	en	32,000	2	123	0	31,875	0.50
BanglaBait	headline	clickbait	bn	15,056	850	124	41	14,041	0.34

et al., 2016); and BanglaBait (Mahtab et al., 2023), 15,056 labelled Bengali news articles with machine-translated English titles. Labels differ in what they assert: LIAR rates individual claims, FakeNewsNet and Twitter15/16 rate a story through a fact-check, and ISOT and WELFake label articles largely by the outlet they came from. We keep these distinctions explicit and draw the strongest conclusions from four *same-task pairs* that hold unit and construct fixed: ISOT↔WELFake (articles), PolitiFact↔GossipCop (headlines, politics vs. entertainment), Twitter15↔Twitter16 (tweets, two collection windows), and LIAR↔PolitiFact (claims vs. headlines from the same fact-checker). Every label is mapped to 1 = fake, false, rumour or clickbait.

Provenance of the article files. For the two article corpora we start from cleaned working copies made at an earlier stage of this project, not from the raw releases, and the difference matters for anyone reproducing the numbers. The ISOT release has 44,898 articles; our copy has 38,833 after exact duplicates and empty texts were removed, which matches the scale of duplication that Verma (2026) reports for this corpus. The WELFake release has 72,134 articles merged from four earlier collections, one of them a Reuters-based news set like ISOT's real class (Verma et al., 2021a,b). Our copy keeps 24,530 articles after duplicates and articles shared with ISOT were removed, so that ISOT↔WELFake transfer is not inflated by shared items; only 18 near-duplicates of ISOT remain, and those are excluded from every test (Section 3.2). The repository documents both files and includes loaders for the original releases, so the whole pipeline, including our own deduplication, can be rerun from the public files.

Table 3: Label-direction probes on the raw files: number of items matching each source signature under raw label 0 / raw label 1. A reliable-outlet signature must sit under "real", a fake-site signature under "fake". ISOT codes 1 = fake; the WELFake file codes 1 = real.

Signature	Expected class	ISOT (0 / 1)	WELFake (0 / 1)
title ends '- The New York Times'	real	0 / 0	0 / 6,223
title ends '- Breitbart'	real	0 / 0	1 / 2,338
text contains '(Reuters)'	real	20,761 / 7	8 / 23
text contains 'Featured image via'	fake	0 / 6,098	125 / 0
title contains 'BREAKING'	fake	8 / 614	199 / 19
text contains '21st Century Wire'	fake	0 / 629	5 / 0
text contains 'Activist Post'	fake	0 / 5	21 / 0

3.2. Audit

Label direction. Corpora disagree on which value denotes "fake", and nothing in a CSV file says which convention it follows. We therefore verify label direction with *signature probes*: strings that only a reliable or only a fabricating outlet produces. Table 3 shows the result for the two article corpora. In ISOT, 20,761 of the 20,768 "(Reuters)" datelines carry label 0 and all 6,098 texts with the fake-site template "Featured image via" carry label 1, so 1 = fake. WELFake is the other way round: all 6,223 titles ending in "- The New York Times" carry label 1 and all 125 "Featured image via" texts carry label 0, in agreement with the official documentation (0 = fake, 1 = real; Verma et al., 2021b). Our working copy carried a class-name column that called label 1 "fake", and our first pipeline trusted it. Read that way, every ISOT↔WELFake score is inverted and ISOT→WELFake looks like a below-chance collapse. The probes exposed the error. A ranking metric below 0.5 on a same-task pair should be treated as a symptom to investigate before it is reported as a finding.

Duplicates. Within each corpus we removed exact duplicates of the normalised text and then near-duplicates found by MinHash locality-sensitive hashing (Broder, 1997) (128 permutations, Jaccard ≥ 0.8; word 3-grams for articles, character 5-grams otherwise). Groups whose members disagree on the label were dropped entirely. Table 2 reports the counts. Two corpora stand out. GossipCop contains 1,174 redundant exact copies of headlines, and a further 356 rows whose headline appears under *both* labels; BanglaBait contains 850 redundant exact copies of titles and, after deduplication, 14,041 items. Even the cleaned WELFake copy still contained 815 near-duplicates.

Cross-corpus overlap. Near-duplicates across corpora inflate transfer. ISOT and WELFake share only 18 near-duplicate articles, so the removal of shared articles from the WELFake copy worked. PHEME, however, shares 58 source tweets with Twitter15 and 128 with Twitter16, because the corpora were collected around

overlapping breaking-news events; Twitter15 and Twitter16 share 169 tweet identifiers, all labelled *unverified*. Every transfer test excludes target items that are near-duplicates of the source corpus.

4. Methodology

4.1. Evaluation protocol

For every corpus and every seed $s \in \{0, \ldots, 4\}$ we draw a stratified 70/10/20 train/validation/test split. The seed also fixes model initialisation and any subsampling. Training splits are capped at 8,000 items, validation at 1,500 and test at 3,000 (stratified) so that the largest corpora do not dominate run time, and every cap is applied in the same way to every model. Featurisers (vocabularies, IDF weights, SVD bases) are fitted on the source training split only. Hyperparameters (the inverse regularisation strength $C \in \{0.25, 1, 4\}$, the number of boosting rounds, or the fine-tuning epoch) are selected on the source validation split by ROC-AUC. The selected model's scores are then Platt-calibrated (Platt, 1999) on the same validation split, which makes the default operating point $t = 0.5$ the Bayes decision under the source prior. No hyperparameter, threshold or calibration step sees a target label. When a model trained on corpus A is tested on corpus B, test items of B that are near-duplicates of any item of A (Section 3.2) are removed.

Three training regimes are run for every model except the fine-tuned transformer. *Full* uses the capped training split. *Matched* subsamples every source to $n = 250$ training items (248 for Twitter16), which removes the size difference between Twitter16 (250 training items) and the article corpora (8,000). *Permuted* shuffles the source training and validation labels; it is a sanity control, and its transfer and in-domain AUC should both be close to 0.5.

4.2. Models

We use seven detectors that span sparse, dense and pretrained representations. Six have a linear or tree-based decision layer, which keeps their behaviour interpretable and their training reproducible on a CPU; the seventh is a fine-tuned transformer: (i) TF-IDF word 1–2-grams with logistic regression (TFIDF-LR), the primary instrument; (ii) the same features with a linear SVM (TFIDF-SVM); (iii) NB-SVM, logistic regression on naive-Bayes-weighted binary n-grams (Wang and Manning, 2012); (iv) averaged GloVe 100-d embeddings (Pennington et al., 2014) with logistic regression (GLOVE-LR); (v) a style model, gradient boosting on 22 surface statistics (length, punctuation and casing rates, digit and stop-word ratios, pronoun rates, weekday mentions, "said" rate, dateline, URL, handle and hashtag rates; STYLE-GB), which sees no vocabulary beyond counts of stop words, pronouns, weekday names and "said", and therefore measures how far a corpus is separable by surface form; (vi) RoBERTa-base (Liu et al., 2019) mean-pooled last-layer representations with logistic regression (ROBERTA-LR), computed on a stratified subset of at most 1,500 items per corpus (first 600 characters); (vii) RoBERTa-base fine-tuned for the task (ROBERTA-FT). Full

fine-tuning was out of reach on our hardware, so the embeddings and the lower eight of the twelve layers are frozen, their outputs are computed once per text and cached, and the upper four layers are trained together with a mean-pooling linear head (AdamW, learning rates 3×10^{-5} for the layers and 10^{-3} for the head, batch 16, three epochs with the epoch chosen on validation AUC, class-weighted loss, at most 128 word pieces for articles and 64 otherwise). It is trained on 800 items per source, validated on 200 and tested on 300 per target, all stratified subsets of the regular splits, for seeds 0–2. For both transformer models, TFIDF-LR is refitted on exactly the same items and characters for a paired comparison. The linear models use scikit-learn (Pedregosa et al., 2011); RoBERTa weights are those packaged in spaCy's `en_core_web_trf` 3.8.0 (Honnibal et al., 2020), fine-tuned with PyTorch.

4.3. Metrics

For every (model, regime, source, target, seed) cell we report threshold-free ranking metrics (ROC-AUC, average precision), threshold-dependent metrics at the default operating point (macro-F1, per-class F1, balanced accuracy, Matthews correlation, predicted-positive rate), calibration (expected calibration error, 15 bins), and macro-F1 at four further operating points described next. Two reference levels help when reading macro-F1: random guessing gives about 0.5, and predicting a single class gives about 0.33. For a single cell, uncertainty is reported as mean ± standard deviation over seeds and as a pooled seed-bootstrap 95% confidence interval (200 resamples of the target test set per seed, pooled over seeds), which captures both split and test-sampling variability. Quantities averaged over the matrix (the ranking share, the recovery rates) depend more on which corpora were collected than on test sampling, and the 42 cells are not independent, since every corpus appears in twelve of them. For these we resample the seven *corpora* with replacement (2,000 draws), keep every directed pair of distinct drawn corpora with its multiplicity, and report the 2.5th and 97.5th percentiles. Paired model comparisons use McNemar's test (McNemar, 1947) per cell and seed; repair comparisons use Wilcoxon signed-rank tests over (target, seed) pairs; families of tests are Holm-corrected (Holm, 1979).

4.4. Transfer gap decomposition

Let $F_{AB}(\tau)$ be the macro-F1 on target B of a model trained on source A, at operating point τ. We consider three operating points on the target scores: the default threshold $\tau_0 = 0.5$; the prior-matched threshold τ_π, which makes the predicted-positive rate equal to the true target prior; and the oracle threshold $\tau^\star = \arg\max_\tau F_{AB}(\tau)$, an upper bound that any thresholding rule could at best reach given the model's ranking of target items. The loss a user experiences when deploying the source model on B, relative to what an in-domain model of B achieves, then decomposes exactly:

$$\underbrace{F_{BB}(\tau_0) - F_{AB}(\tau_0)}_{\text{transfer gap}} = \underbrace{\big[F_{BB}(\tau_0) - F_{AB}(\tau^\star)\big]}_{\text{ranking deficit}} + \underbrace{\big[F_{AB}(\tau^\star) - F_{AB}(\tau_\pi)\big]}_{\text{threshold residual}} + \underbrace{\big[F_{AB}(\tau_\pi) - F_{AB}(\tau_0)\big]}_{\text{prior / calibration loss}}. \quad (1)$$

No choice of threshold can recover the first term. It measures how much worse the source model *orders* target items than the target's own model does. The last two terms are *operating-point* losses and would vanish with the right threshold. Adding $F_{AA}(\tau_0) - F_{BB}(\tau_0)$ gives the gap relative to the source's own in-domain score, which is what a benchmark user is promised; this difficulty term averages to zero over a full matrix and is reported per cell only.

Two details of Eq. (1) could bias the split, and we check both. First, the ranking term compares the transfer oracle with the in-domain model at its *default* threshold, so it also contains the in-domain threshold slack $F_{BB}(\tau^\star_{BB}) - F_{BB}(\tau_0)$. Second, $\tau^\star$ is chosen on the same items it is scored on, which flatters $F_{AB}(\tau^\star)$ on small test sets and so inflates the operating-point share. We report the slack, and we recompute the decomposition with a *cross-fitted* oracle, whose threshold is tuned on a random half of the target test set and scored on the other half (five repeats, both halves). We also repeat the decomposition with balanced accuracy, which, unlike macro-F1, does not depend on the class prior.

4.5. Label-free operating points

We compare four ways of choosing a threshold without target labels. (a) *EM prior adaptation* (Saerens et al., 2002) re-estimates the target prior from the source-calibrated posteriors by expectation–maximisation, adjusts the posteriors and thresholds them at 0.5. With Platt-calibrated scores this is the bias-corrected recipe recommended by Alexandari et al. (2020), and it is optimal under pure label shift. (b) *Rate transfer* sets the threshold so that the share of target items flagged equals the source base rate π_A. It transfers the model's ranking-based decision rule rather than its score scale. We also report the variant that flags the share q_A the source model flags on its own held-out test split. (c) *Adjusted classify-and-count* (Forman, 2008) estimates the target prior as $\hat{\pi}_B = (\hat{r}_B - \mathrm{FPR}_A)/(\mathrm{TPR}_A - \mathrm{FPR}_A)$, where $\hat{r}_B$ is the share of target items above 0.5 and the error rates come from the source test split, and then flags the top $\hat{\pi}_B$ of target items. (d) A *two-component Gaussian mixture* on the target logits places the threshold where the components' posteriors cross. As references, we report the share 0.5, which needs no information at all, and the true target prior π_B, which needs one number about the target but no item labels.

Rate transfer is a quantile threshold, and quantile thresholds are an old idea in quantification (Forman, 2008). We do not claim the rule as new. What matters here is *when* it should beat EM, and that can be stated exactly. Let F^A_y and F^B_y be the distributions of the source model's scores for class y on source and target.

Proposition 1. *Suppose (i) there is a strictly increasing map g with $F^B_y(g(u)) = F^A_y(u)$ for both classes and all u, and (ii) $\pi_B = \pi_A$. Then flagging the top q_A share of target items reproduces on B the true- and false-positive rates that the threshold τ_0 has on A, whereas the fixed threshold τ_0 does so only if $g(\tau_0) = \tau_0$. If instead g is the identity and $\pi_B \neq \pi_A$ (pure label shift), EM prior adaptation with calibrated scores is consistent and rate transfer is not.*

The proof is in Appendix A. The two rules thus suit two different kinds of shift. EM assumes the score distributions within each class stay fixed while the class balance changes. Rate transfer assumes the reverse: the class balance stays roughly fixed while the scores are distorted by an order-preserving map. When the distortion is not order-preserving, part of the loss becomes a ranking deficit that no threshold can recover, which is the first term of Eq. (1). Section 5.4 checks which assumption our corpora come closer to, and a controlled experiment maps what happens when (ii) fails: each target test set is resampled (1,000 items, three draws) to a base rate of $\pi_A + \delta$ for $\delta \in \{-0.3, \ldots, 0.3\}$, clipped to $[0.05, 0.95]$, and every operating point is recomputed.

4.6. Mechanism probes

To identify what signal carries in-domain performance, we use eight probes, each run with TFIDF-LR unless stated: *surface statistics only* (STYLE-GB above); *group-only*, out-of-fold smoothed target encoding of group identity (publisher domain for PolitiFact, GossipCop and BanglaBait; speaker for LIAR; event for PHEME), i.e. how well the label is predicted by *who* published an item; *learning curves* from 32 training items to the full split; *top-K deletion*, removing the $K \in \{10, \ldots, 3000\}$ highest-$|w|$ features and retraining (cf. Verma, 2026); *masking* of source tags and datelines, of time expressions and numbers, of named entities (spaCy `en_core_web_sm`), or of all three; *topic-balanced training*, where classes are subsampled to equal size inside each of 50 k-means topic clusters, removing topic–label correlation from the training set; *corpus separability*, the proxy $\mathcal{A}$-distance $2(1-2\epsilon)$ between two corpora, with ϵ the cross-validated error of a classifier that tells them apart from text alone (Ben-David et al., 2010); and *feature attribution*, the largest-weight n-grams per class and their linear-SHAP magnitudes $|w_j|\,\mathbb{E}|x_j - \bar{x}_j|$. For the more expensive probes (top-K, masking, vocabulary ablation) training is capped at 4,000 and test at 1,500 items; masking and vocabulary ablation use three seeds.

4.7. Representation repairs

The ranking deficit can only be reduced by a better representation. We test repairs in a leave-one-corpus-out (LOCO) setting: for each held-out veracity corpus, the model is trained on the union of the other six (each capped at 1,500 training and 300 validation items) and evaluated on the held-out test split, with near-duplicates of any training corpus removed. Methods are: *pooled* training; pooled training with sample weights that equalise every (corpus, class) cell (*balanced*); balanced training on texts with source tags, datelines, URLs, handles and time expressions *masked*; balanced training after deleting the 2,000 features most predictive of *corpus identity* (from a one-vs-rest corpus classifier); and three dense variants on a 300-d SVD of the TF-IDF space: the SVD baseline, *INLP* (Ravfogel et al., 2020), which projects out 15 iterations of linear corpus-identity directions, and *CORAL* (Sun et al., 2016), which aligns source feature covariance to the (unlabelled) target. None uses target labels; CORAL alone uses unlabelled target text.

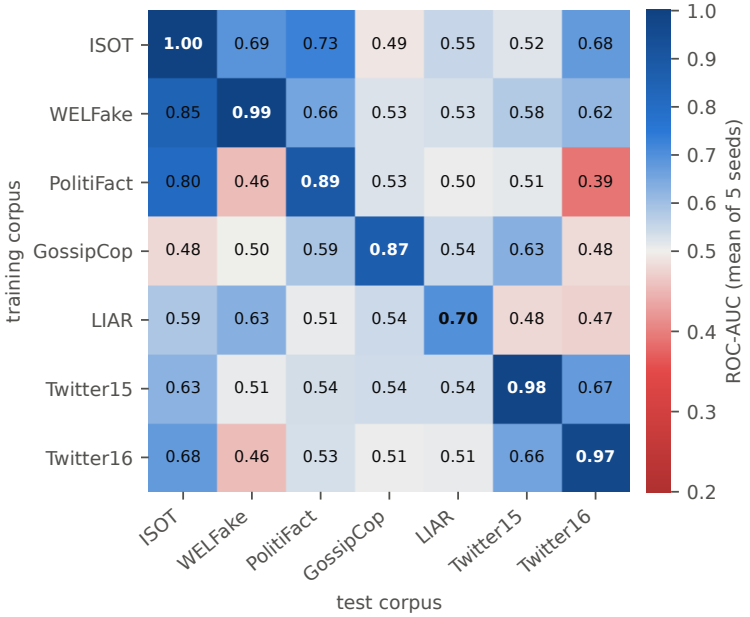


Figure 1: Transfer ROC-AUC of TFIDF-LR (mean of five seeds). Rows are training corpora, columns test corpora; the diagonal is in-domain. Blue is above chance, red below; the midpoint is 0.5.

5. Results

5.1. The size of the gap

Table 4 and Figure 1 summarise the seven-corpus veracity matrix. With TFIDF-LR, mean in-domain ROC-AUC is 0.914, mean off-diagonal AUC is 0.567, and on the eight same-task cells it is 0.624. The sanity controls behave as they should. With permuted labels, transfer AUC is 0.502 ($\pm$0.067 across cells and seeds) and in-domain AUC 0.514. With every source cut to 250 training items the gap remains (in-domain 0.861, off-diagonal 0.556), so it does not come from the very different corpus sizes.

The gap is real, but it is not a collapse to chance. On the cleanest pair, ISOT$\rightarrow$WELFake, AUC is 0.69 [0.66, 0.71], and in the reverse direction 0.85 [0.82, 0.87] (Table 5). Two pairs carry almost no transferable ranking: PolitiFact$\rightarrow$LIAR (0.50 [0.47, 0.54]) and LIAR$\rightarrow$PolitiFact (0.51 [0.42, 0.60]). Both are labelled by the same fact-checker, but one rates claims and the other news stories. In 8 of the 42 off-diagonal cells (19%) the mean AUC is below 0.5, meaning that the cue a source teaches points the wrong way on that target.

Threshold-dependent metrics look much worse than AUC. Off-diagonal macro-F1 at the default threshold averages 0.439, not far above the one-class level of 0.33, against 0.854 in-domain. The predicted-positive rates in Table 5 show why. An ISOT model flags 0.78 of WELFake as fake although the fake rate is 0.43, and a GossipCop model flags only 0.07 of PolitiFact although its fake rate is 0.43. Scores that were well

Table 4: In-domain versus transfer performance on the seven veracity corpora (mean ± s.d. over seeds of the per-seed mean over cells). "Pairs" averages the eight directed same-task cells. Off-diagonal macro-F1 is given at the default threshold and at the oracle threshold. ROBERTA-LR is evaluated on the 1,500-item subsets and ROBERTA-FT on 800 training and 300 test items per corpus with three seeds (Section 4.2); compare each with its paired TFIDF-LR in Section 5.5, not with the rows above.

		ROC-AUC			macro-F1		
Model	Regime	In-domain	Off-diag.	Pairs	In-domain	Off-diag.	Off-diag. oracle
TFIDF-LR	full	0.914 ±0.006	0.567 ±0.006	0.624 ±0.004	0.854 ±0.005	0.439 ±0.011	0.565 ±0.004
TFIDF-LR	matched	0.861 ±0.010	0.556 ±0.004	0.590 ±0.009	0.786 ±0.012	0.446 ±0.014	0.558 ±0.002
TFIDF-LR	permuted	0.519 ±0.026	0.499 ±0.012	0.514 ±0.019	0.397 ±0.014	0.377 ±0.015	0.518 ±0.007
TFIDF-SVM	full	0.914 ±0.005	0.564 ±0.005	0.625 ±0.005	0.855 ±0.005	0.437 ±0.009	0.564 ±0.004
TFIDF-SVM	matched	0.862 ±0.010	0.554 ±0.005	0.592 ±0.010	0.787 ±0.011	0.447 ±0.017	0.556 ±0.003
TFIDF-SVM	permuted	0.519 ±0.024	0.498 ±0.015	0.511 ±0.020	0.398 ±0.009	0.377 ±0.012	0.516 ±0.008
NB-SVM	full	0.912 ±0.004	0.547 ±0.006	0.590 ±0.011	0.853 ±0.008	0.410 ±0.008	0.555 ±0.003
NB-SVM	matched	0.862 ±0.011	0.540 ±0.006	0.559 ±0.004	0.786 ±0.012	0.398 ±0.013	0.549 ±0.004
NB-SVM	permuted	0.510 ±0.023	0.494 ±0.009	0.502 ±0.009	0.400 ±0.003	0.383 ±0.012	0.520 ±0.006
GLOVE-LR	full	0.864 ±0.005	0.564 ±0.005	0.610 ±0.005	0.782 ±0.007	0.465 ±0.004	0.564 ±0.004
GLOVE-LR	matched	0.830 ±0.006	0.563 ±0.010	0.592 ±0.017	0.740 ±0.011	0.466 ±0.010	0.564 ±0.007
GLOVE-LR	permuted	0.520 ±0.022	0.504 ±0.026	0.494 ±0.021	0.404 ±0.020	0.391 ±0.012	0.525 ±0.019
STYLE-GB	full	0.745 ±0.009	0.535 ±0.004	0.595 ±0.008	0.679 ±0.005	0.427 ±0.007	0.548 ±0.003
STYLE-GB	matched	0.702 ±0.008	0.526 ±0.009	0.561 ±0.032	0.632 ±0.008	0.413 ±0.017	0.542 ±0.007
STYLE-GB	permuted	0.503 ±0.018	0.502 ±0.016	0.493 ±0.028	0.410 ±0.008	0.390 ±0.007	0.523 ±0.011
ROBERTA-LR	full	0.861 ±0.012	0.544 ±0.008	0.630 ±0.011	0.777 ±0.013	0.452 ±0.013	0.549 ±0.010
ROBERTA-FT	full	0.867 ±0.010	0.559 ±0.013	0.614 ±0.020	0.795 ±0.005	0.418 ±0.009	0.571 ±0.008

calibrated in-domain are badly calibrated on the target.

5.2. Where the gap goes

Applying Eq. (1) cell by cell (Table 6, Figure 2) splits the loss into a part that no threshold can recover and a part that the right threshold would. For TFIDF-LR on the same-task pairs, the mean transfer gap is 0.31 macro-F1 points, of which 0.24 (79%) is a ranking deficit and the rest an operating-point loss. On the other cells the gap is 0.44, and 68% of it is ranking. Over all cells, 70% of the loss is ranking (corpus-bootstrap 95% interval [63%, 79%]) and 30% is operating point. Most of the gap is ranking, then, and part of it is threshold. Both parts matter. The operating-point share is large enough to change which of two detectors looks better, and it is the cheaper one to fix.

Table 5: Same-task pairs (TFIDF-LR, five seeds). AUC with pooled seed-bootstrap 95% CI; "In" is the source's in-domain AUC; prior is the target fake rate and PPR the predicted-positive rate at $t = 0.5$; macro-F1 at the default, prior-matched and oracle thresholds; ECE of the source-calibrated scores on the target.

Pair	In	AUC [95% CI]	Prior	PPR	$F1_{0.5}$	$F1_{\pi}$	$F1^{\star}$	ECE
ISOT→WELFake	0.999	0.690 [0.66, 0.71]	0.43	0.78	0.539	0.635	0.638	0.38
WELFake→ISOT	0.989	0.847 [0.82, 0.87]	0.46	0.36	0.753	0.763	0.765	0.11
PolitiFact→GossipCop	0.893	0.526 [0.48, 0.56]	0.23	0.54	0.464	0.514	0.517	0.32
GossipCop→PolitiFact	0.869	0.589 [0.50, 0.68]	0.43	0.07	0.425	0.576	0.595	0.28
Twitter15→Twitter16	0.981	0.672 [0.52, 0.82]	0.46	0.62	0.572	0.597	0.646	0.25
Twitter16→Twitter15	0.967	0.657 [0.55, 0.75]	0.51	0.63	0.580	0.591	0.631	0.28
LIAR→PolitiFact	0.698	0.508 [0.42, 0.60]	0.44	0.54	0.495	0.517	0.536	0.12
PolitiFact→LIAR	0.893	0.505 [0.47, 0.54]	0.44	0.40	0.497	0.502	0.509	0.22

5.3. How robust is the split?

The split survives every check we ran. Dropping any one corpus from the matrix leaves the ranking share between 67% and 72%. The in-domain threshold slack that the ranking term absorbs is small, 0.014 macro-F1 on average (at most 0.037). The in-sample oracle is optimistic, as expected: a threshold tuned on one half of the target and scored on the other reaches 0.537, which is 0.028 below the in-sample oracle on average and 0.056 below it on the targets with fewer than 250 test items. With the cross-fitted oracle the ranking share rises to 76%, so if anything Eq. (1) overstates the operating-point part. With balanced accuracy in place of macro-F1, the ranking share is 84%. Across detectors the ranking share varies more (Table 6): it is lowest for the style model on the non-paired cells and for the fine-tuned transformer (59% overall; Section 5.5), but ranking is the larger part for every detector.

5.4. Repairing the operating point without labels

Table 7 and Figure 3 compare operating points. EM prior adaptation, the textbook remedy, *lowers* off-diagonal macro-F1 from 0.439 to 0.358. It recovers −64% of the operating-point loss (interval [−154%, −27%]), and it does so for every detector (−87% to −41%). Rate transfer, which flags the same share of target items as the source base rate, raises macro-F1 to 0.541 and recovers 81% ([56%, 91%]), close to the 86% reached when the true target base rate is known (0.547). It recovers between 66% and 81% across the seven detectors, and it is at least as good as the cross-fitted oracle, that is, as a threshold tuned on half of the labelled target set (recovery 104% against that oracle). Flagging the share q_A that the source model flags in-domain recovers 76%, and the mixture model 57%. Rate transfer is worse than the default threshold in 10% of the cell-seed combinations.

Table 6: Transfer gap decomposition (Eq. 1), macro-F1 points, mean over cells and seeds. The gap is relative to the target's own in-domain score.

Model	Cells	Gap	Ranking	Threshold	Prior/cal.	Ranking share
TFIDF-LR	pairs	0.308	0.244	0.018	0.046	79%
TFIDF-LR	other cells	0.441	0.300	0.018	0.122	68%
TFIDF-SVM	pairs	0.308	0.243	0.016	0.049	79%
TFIDF-SVM	other cells	0.444	0.303	0.018	0.123	68%
NB-SVM	pairs	0.319	0.261	0.023	0.036	82%
NB-SVM	other cells	0.472	0.306	0.021	0.145	65%
GLOVE-LR	pairs	0.281	0.193	0.016	0.071	69%
GLOVE-LR	other cells	0.327	0.224	0.017	0.085	69%
STYLE-GB	pairs	0.141	0.091	0.013	0.036	65%
STYLE-GB	other cells	0.278	0.141	0.020	0.117	51%
ROBERTA-LR	pairs	0.243	0.162	0.022	0.059	66%
ROBERTA-LR	other cells	0.344	0.244	0.024	0.076	71%
ROBERTA-FT	pairs	0.340	0.186	0.031	0.123	55%
ROBERTA-FT	other cells	0.387	0.233	0.027	0.127	60%

Two observations keep these numbers in proportion. First, flagging exactly half of the items recovers 80%, almost as much as rate transfer. The source base rates of our corpora lie between 0.23 and 0.51, so in this benchmark most of the gain comes from switching from a score threshold to *any* sensible rate threshold, and only a small part from knowing the source rate. Second, the source base rate is a good guess of the target base rate here (mean absolute error 0.09), which is condition (ii) of Proposition 1.

Why does EM fail? The answer is in the calibration. EM is optimal when only $p(y)$ changes and calibrated posteriors stay meaningful. Here calibration does not survive the move: expected calibration error is 0.046 in-domain but 0.328 on the target. EM's prior estimates therefore drift towards 0 or 1 (mean absolute error 0.46), far worse than simply assuming the source prior (0.09). Adjusted classify-and-count fails for the same reason, because the source error rates it inverts do not hold on the target either (prior error 0.41, recovery -27%). The shift is in $p(x \mid y)$, that is, in style, topic and source, much more than in $p(y)$. That is the regime in which Proposition 1 favours transferring the rate over transferring the scores.

When the base rate moves. Condition (ii) will not always hold, so we resampled every target to base rates between 0.3 below and 0.3 above the source rate (Figure 4). Rate transfer degrades as the shift grows: it recovers 81% of the operating-point loss with no shift, 58% at -0.3 and 22% at $+0.3$, where its macro-F1 (0.491) falls well below that of a threshold set at the true rate (0.541). It still beats the default threshold

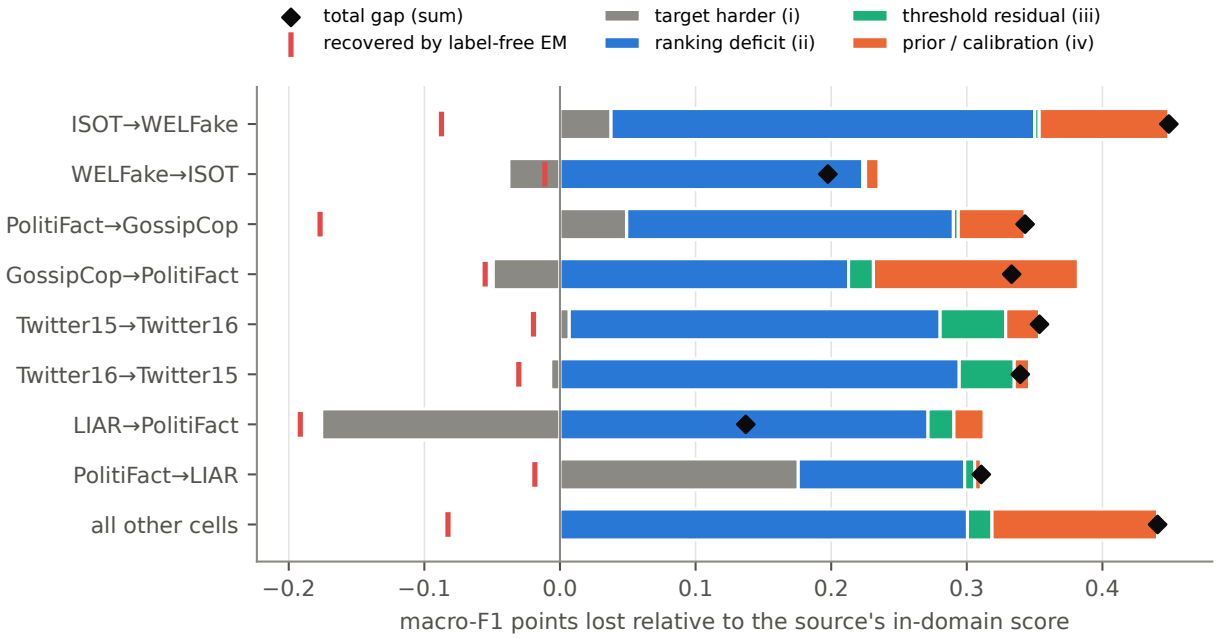


Figure 2: Transfer gap decomposition for TFIDF-LR (macro-F1 points relative to the source's in-domain score, mean of five seeds). Grey: target harder or easier than source; blue: ranking deficit; green and orange: operating-point losses. Diamonds mark the total; red ticks show what EM prior adaptation recovers (negative values make things worse).

at every shift we tried, by 0.02 to 0.11 macro-F1. The two label-free prior estimators do not help: EM and adjusted classify-and-count stay below the default threshold at every shift, because both rely on score distributions that do not transfer. In practice, then, rate transfer is a safe default, but when the base rate of the new feed is expected to differ from the training data, an estimate of that rate from a small labelled sample is worth more than any label-free correction we tested.

5.5. The gap across models

The picture does not depend on the detector (Table 4; full matrices for every model are in the Supplementary Material). The four content models reach in-domain AUC between 0.864 and 0.914 and off-diagonal AUC between 0.547 and 0.567, and all of them show the same split between ranking and operating-point loss (Table 6). Averaged word embeddings (GLOVE-LR) are weaker in-domain (0.864) and transfer no better (0.564).

The style model is the most revealing. With only 22 surface statistics it reaches an average in-domain AUC of 0.745 (0.999 on ISOT, 0.951 on WELFake), and its transfer AUC (0.535) is only a few points below that of the lexical models. A large part of what these benchmarks reward is form: length, casing, punctuation and attribution habits (Section 5.6).

Pretrained encoders do not change the conclusion. On identical 1,500-item subsets, frozen RoBERTa

Table 7: Off-diagonal macro-F1 under different operating points (mean over the 42 cells and five seeds). Label-free: default $t = 0.5$, EM prior adaptation, rate transfer (predicted-positive rate set to the source base rate), predicted-positive rate 0.5, and a 2-component GMM on target logits. With one number known: rate set to the true target prior. Upper bound: oracle threshold. For TFIDF-LR, Holm-corrected Wilcoxon p versus default: EM 1.6e-32, rate transfer 6.6e-31, rate 0.5 1.4e-29, GMM 1.7e-18.

	label-free					prior known	bound
Model	$t = 0.5$	EM	rate transfer	rate 0.5	GMM	rate $= \pi_B$	oracle
TFIDF-LR	0.439	0.358	0.541	0.539	0.510	0.547	0.565
TFIDF-SVM	0.437	0.363	0.539	0.537	0.503	0.546	0.564
NB-SVM	0.410	0.349	0.525	0.526	0.444	0.534	0.555
GLOVE-LR	0.465	0.381	0.540	0.542	0.494	0.547	0.564
STYLE-GB	0.427	0.347	0.521	0.521	0.477	0.529	0.548
ROBERTA-LR	0.452	0.368	0.516	0.516	0.477	0.525	0.549
ROBERTA-FT	0.418	0.340	0.535	0.534	0.479	0.544	0.571

features give in-domain AUC 0.861 and off-diagonal AUC 0.544, against 0.895 and 0.549 for TFIDF-LR refitted on the same splits. The paired off-diagonal difference is −0.005 AUC (Wilcoxon $p = 0.44$), and RoBERTa is better in 17 of the 42 cells. Fine-tuning the upper four layers does not change this either. On the same 800 training and 300 test items per corpus (3 seeds), ROBERTA-FT reaches in-domain AUC 0.867 and off-diagonal AUC 0.559, against 0.892 and 0.558 for the paired TFIDF-LR; the off-diagonal difference is +0.001 (Wilcoxon $p = 0.98$, better in 23 of 42 cells). Its gap is again mostly ranking (59%), EM again makes the operating point worse (−51% recovered), and rate transfer again recovers most of it (77%). With only four layers trained on 800 items, the fine-tuned model is slightly weaker than TFIDF-LR in-domain (−0.025 AUC), so this is not the strongest transformer one could build; but its transfer matches that of the lexical model, and the shape of its gap is the same. This agrees with Verma (2026) and Dell'Oglio et al. (2026), who found that fully fine-tuned transformers gain in-domain but not, or not reliably, across corpora. Pairwise McNemar tests on the same target items (Holm-corrected over all comparisons) find significant differences from TFIDF-LR in the following share of (cell, seed) comparisons: GLOVE-LR 36%, NB-SVM 21%, ROBERTA-LR 17%, STYLE-GB 47%, TFIDF-SVM 12%.

5.6. What the detectors learn

Cheap signals go a long way. Table 8 lists four shortcut indicators per corpus. The surface-statistics model recovers on average 57% of the above-chance in-domain AUC of the full lexical model, and on the article corpora almost all of it (ISOT 0.999, WELFake 0.951). Sixty-four training examples are enough for an in-domain AUC of 0.983 on ISOT and 0.894 on WELFake; the claim and celebrity-headline corpora, whose

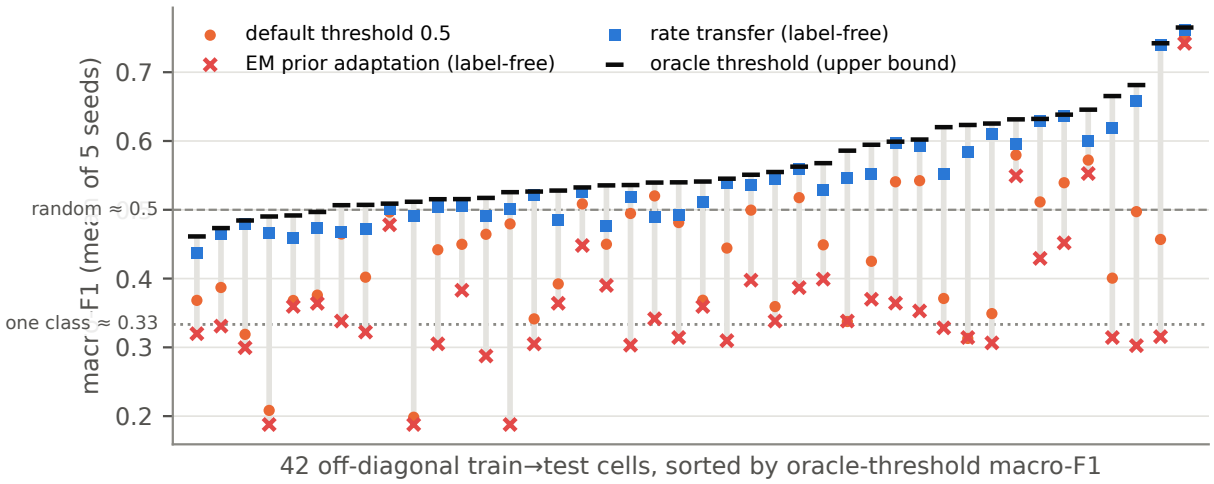


Figure 3: Off-diagonal macro-F1 of TFIDF-LR for each of the 42 cells (mean of five seeds), sorted by the oracle bound. Rate transfer closes most of the distance from the default threshold to the oracle without any target label; EM prior adaptation often falls below the default.

in-domain scores are lower to begin with, learn much more slowly (LIAR 0.567, GossipCop 0.610). On the headline and claim corpora, publisher or speaker identity alone predicts the label with AUC 0.840 (PolitiFact), 0.807 (GossipCop) and 0.664 (LIAR). When a label can be predicted from form, from a few dozen examples or from who published an item, a high in-domain score says little about veracity.

The strongest cues are collection signatures. Figure 5 shows the highest-weight n-grams. In ISOT, the real class is signalled by Reuters wire conventions ("reuters", "said", "washington reuters", weekday attributions such as "on thursday") and the fake class by the template of the sites it was collected from ("featured image", "image via", "read more", "video"). In WELFake, the real class is signalled by publisher suffixes inherited from one of its source collections ("breitbart", "york times", "mr") and the fake class by the window in which fake items were gathered ("2016", "october", "november", "hillary"). Each corpus teaches its own proxy, which is why the sign of a cue can flip from one corpus to the next (Section 5.1).

Removing cues does not remove the signal. Deleting the top-K features and retraining barely moves in-domain AUC: on ISOT it goes from 0.999 with all features to 0.991 after deleting 1,000 and 0.985 after 3,000, and on WELFake from 0.984 to 0.962 and 0.949; transfer to the paired corpus, in contrast, falls from 0.673 to 0.580 (ISOT→WELFake) and from 0.827 to 0.718 (WELFake→ISOT). So the highest-weight features carry most of what transfers, while the signal that keeps in-domain AUC near its ceiling is spread over thousands of corpus-specific features. Masking source tags, datelines, time expressions, numbers and named entities together (Table 9) leaves in-domain AUC at 0.996 (ISOT) and 0.964 (WELFake), and moves ISOT→WELFake from 0.679 to 0.650 and WELFake→ISOT from 0.808 to 0.810. Balancing classes within each of 50 topic

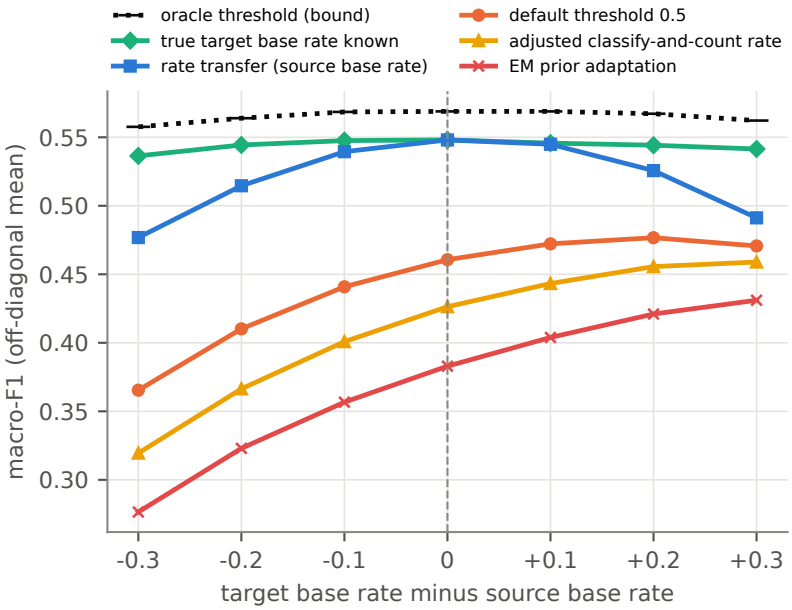


Figure 4: Off-diagonal macro-F1 of TFIDF-LR when every target test set is resampled to the source base rate plus a shift (42 cells, five seeds, three draws of 1,000 items). Rate transfer is best when the base rate is unchanged and degrades as it moves; the label-free prior estimators (EM, adjusted classify-and-count) stay below the default threshold throughout.

clusters, whose class purity averages 0.82 in ISOT and 0.81 in WELFake, leaves in-domain AUC unchanged (ISOT 0.998→0.999, WELFake 0.984→0.986) and transfer unchanged (0.604→0.606, 0.622→0.622); only GossipCop loses in-domain AUC (0.846→0.799). The separating signal is diffuse. It is spread through the style and register of the whole text rather than sitting in a list of tokens or topics that could be removed, as Verma (2026) also found for ISOT.

Corpus divergence does not predict transfer. Every pair of corpora can be told apart from text alone (proxy $\mathcal{A}$-distance between 1.28 and 1.98, maximum 2; ISOT–WELFake 1.45), yet across the 42 cells we find no association between source–target divergence and transfer AUC (Spearman $\rho = -0.01$). With seven corpora the corpus-bootstrap 95% interval is wide ([-0.61, 0.69]), so a moderate association cannot be excluded, but label-free divergence is clearly not a usable predictor of transfer here. What matters for transfer is less how different two corpora look than whether the cue that separates the classes in one also separates them, with the same sign, in the other.

5.7. Repairing the ranking

Table 10 reports the leave-one-corpus-out repairs (per-target values in Table 11). Training on the six other corpora at once is the only change that helps ranking. Held-out AUC rises to 0.660, against 0.567 for the average single source. That is slightly below the best single source chosen with hindsight (0.671),

Table 8: Shortcut indicators (ROC-AUC, mean of five seeds). In-domain AUC of TFIDF-LR with the full training split, of the surface-statistics STYLE-GB, and of TFIDF-LR trained on only 64 or 256 items; AUC of a group-identity baseline (publisher domain for PolitiFact and GossipCop, speaker for LIAR); and the mean off-diagonal AUC with the corpus as source and as target.

	in-domain					transfer	
Corpus	full	style only	$n = 64$	$n = 256$	group only	as source	as target
ISOT	0.999	0.999	0.983	0.991	–	0.611	0.670
WELFake	0.989	0.951	0.894	0.941	–	0.626	0.540
PolitiFact	0.893	0.787	0.742	0.850	0.840	0.533	0.592
GossipCop	0.869	0.659	0.610	0.723	0.807	0.538	0.522
LIAR	0.698	0.590	0.567	0.607	0.664	0.537	0.530
Twitter15	0.981	0.603	0.793	0.962	–	0.572	0.563
Twitter16	0.967	0.625	0.890	–	–	0.556	0.552

which cannot be identified without target labels, and far below the targets' own in-domain AUC of 0.914. Nothing we added on top of pooling helped on average. Balancing every (corpus, class) cell (0.659, Δ-0.001, $p = 1.000$) and masking collection signatures (0.661, Δ+0.001, $p = 1.000$) leave AUC unchanged on average; masking helps clearly only on ISOT (Table 11), the corpus whose signatures it was written for.

Methods that make the representation *invariant to corpus identity* make ranking worse: deleting the 2,000 most corpus-predictive features gives 0.636 (Δ-0.023, $p = 0.006$), CORAL 0.621 (Δ-0.038, $p < 0.001$) and INLP 0.605 (Δ-0.054, $p = 0.001$). INLP does what it is designed to do, reducing linear corpus-identity accuracy from 0.87 to 0.25 (six classes), but it removes veracity information with it. In these benchmarks corpus identity and label are entangled. Each corpus collected its fake and real items from different outlets, periods and formats, so the directions that identify a corpus are partly the directions that separate its classes. Making a model blind to the corpus does not make it blind to the shortcut. This is also the assumption behind domain-adversarial and domain-gated detectors (Wang et al., 2018; Nan et al., 2021), and on these corpora it does not hold. The ranking gap that remains looks like a property of the data more than of the learner.

5.8. A low-resource language: Bengali clickbait

Replication and duplicates. On the official BanglaBait split, character 3–5-gram logistic regression reaches a clickbait-class F1 of 64.37 (Table 12), within two points of the 66.28 reported for the same model by Mahtab et al. (2023), and a linear SVM reaches 63.79. Removing the near-duplicates found by MinHash from the official splits lowers the same model to 59.15 (AUC 0.834→0.823): even after the original authors' Levenshtein filter, duplicated titles inflate the benchmark. Over five random re-splits of the deduplicated

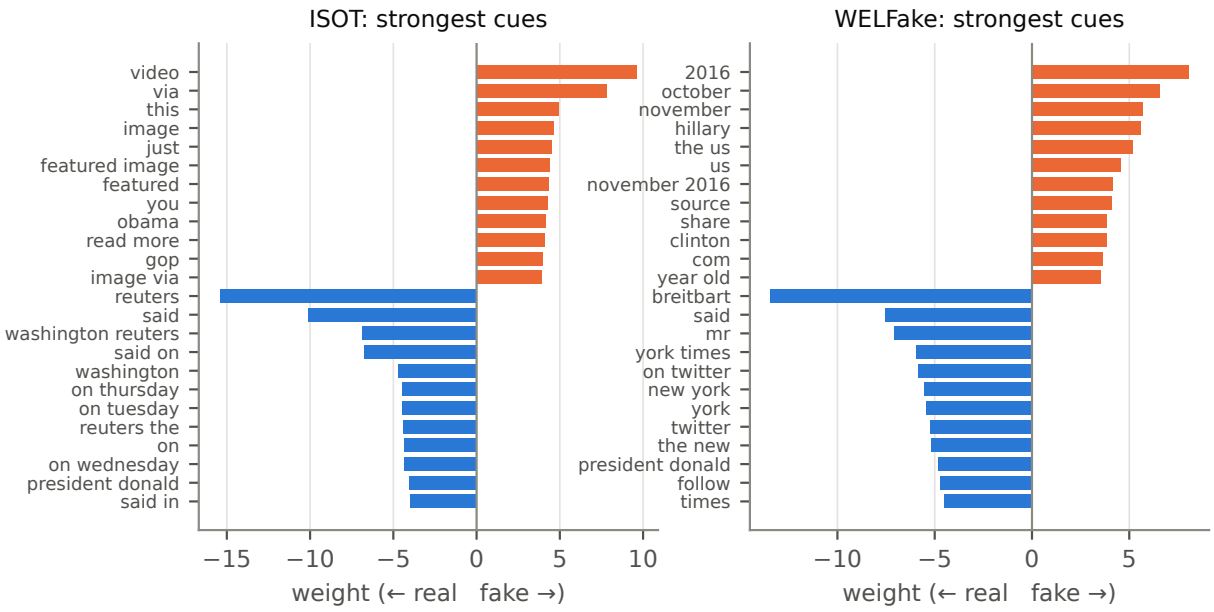


Figure 5: Highest-weight n-grams of TFIDF-LR (seed 0) for the real (blue) and fake (orange) classes. Both corpora are separated mainly by outlet conventions and collection-period vocabulary.

corpus the model reaches AUC 0.821 ± 0.011.

Publisher identity. The 19 publishers with at least 100 articles have clickbait rates from 7% to 66%, so publisher identity alone predicts the label with AUC 0.687 (Section 5.6). Leave-one-publisher-out evaluation nevertheless costs little: across 20 held-out publishers (the smallest ones pooled), mean AUC is 0.791 (range 0.651–0.911), close to the random-split value. Bengali clickbait detection survives a change of source that English veracity detection does not. One reading is that this is a construct effect rather than a language effect. Clickbait *is* a property of a headline's style, so a style-sensitive detector can learn the construct itself, whereas for veracity, style is only a correlate of the label that changes from corpus to corpus. The comparison is not fully controlled, though: BanglaBait was labelled by annotators reading each headline, while the English veracity labels come from fact-checks or from the outlet an article was taken from, so how the labels were made differs as well as what they mean.

Across languages. A translate-test probe uses BanglaBait's machine-translated English titles. A detector trained on English clickbait headlines (Chakraborty et al., 2016) ranks translated Bengali titles with AUC 0.707, against 0.777 for a detector trained on the translated titles themselves and 0.813 for one trained on the original Bengali; translation costs about 0.04 AUC and most of the in-language ranking ability carries over. In the other direction, the translated-Bengali detector reaches 0.931 on English headlines (in-domain 0.994). Clickbait carries over across languages much better than veracity carries over across English corpora,

Table 9: Masking ablation (TFIDF-LR, ROC-AUC, three seeds, first 3,000 characters). Rows: training corpus and what is masked in every corpus; columns: test corpus (the training corpus's own column is in-domain).

Train	Masked	ISOT	WELFake	PolitiFact	GossipCop	LIAR
ISOT	none	0.998	0.679	0.739	0.481	0.559
ISOT	source tags, datelines, URLs, handles	0.997	0.681	0.733	0.483	0.559
ISOT	dates, weekdays, numbers	0.998	0.692	0.723	0.479	0.563
ISOT	named entities	0.996	0.657	0.678	0.543	0.557
ISOT	all of the above	0.996	0.650	0.678	0.540	0.556
WELFake	none	0.808	0.980	0.636	0.529	0.532
WELFake	source tags, datelines, URLs, handles	0.815	0.979	0.634	0.533	0.536
WELFake	dates, weekdays, numbers	0.831	0.977	0.608	0.509	0.526
WELFake	named entities	0.833	0.965	0.614	0.511	0.522
WELFake	all of the above	0.810	0.964	0.611	0.500	0.522

although translation and the change of collection still cost about 0.1 AUC (0.707 against 0.813 in-language).

6. Discussion

What the benchmarks measure. Every corpus in our suite can be told apart from every other with high accuracy (proxy $\mathcal{A}$-distance 1.28–1.98 on a scale whose maximum is 2). On the article corpora a model of surface statistics recovers nearly all of the in-domain signal, and on average more than half of it. The article and tweet corpora reach most of their final AUC with 64 training examples, and the heaviest feature weights point to outlet signatures and to the time window in which the fake class was collected (Section 5.6). To a large extent, then, in-domain scores measure how well a model recognises the *collection process* behind each class. Verma (2026) reached the same conclusion for ISOT alone. Our matrix shows that it holds across units and constructs, and that what is lost on a new corpus is mostly ranking, with calibration a smaller part.

What to do about it. The decomposition separates a cheap problem from a hard one. The operating-point part is cheap. Rate transfer needs no target labels and one line of code, and it recovered 81% of that loss for TFIDF-LR and between 66% and 81% across detectors, while the textbook label-shift correction made things worse for all of them. Someone deploying a detector on a new feed should therefore fix the *share* of items they flag, not a score threshold. If the base rate of the feed is expected to differ from the training data, labelling a small sample to estimate it is the better investment, since neither label-free prior estimator we tested could do it (Section 5.4).

Table 10: Leave-one-corpus-out repairs (TFIDF-LR features; mean over the seven held-out veracity corpora and five seeds; s.d. across seeds). ΔAUC versus pooled training with Holm-corrected Wilcoxon p over the 35 (target, seed) pairs. For reference, the mean single-source transfer AUC to the same targets is 0.567, the best single source (chosen with target labels) 0.671, and the targets' own in-domain AUC 0.914.

Training	AUC	AP	$F1_{0.5}$	$F1^{\star}$	ΔAUC (p)
pooled (6 corpora)	0.660 ±0.013	0.582	0.581	0.634	–
+ (corpus, class) balancing	0.659 ±0.013	0.583	0.582	0.632	-0.001 (1.000)
+ balancing + masking	0.661 ±0.012	0.584	0.576	0.633	+0.001 (1.000)
+ balancing + corpus-feature deletion	0.636 ±0.011	0.556	0.560	0.611	-0.023 (0.006)
SVD-300 + balancing	0.647 ±0.009	0.566	0.583	0.624	-0.013 (0.043)
SVD-300 + INLP	0.605 ±0.011	0.520	0.525	0.591	-0.054 (0.001)
SVD-300 + CORAL	0.621 ±0.008	0.539	0.563	0.604	-0.038 (<0.001)

Table 11: Leave-one-corpus-out AUC per held-out corpus (mean of five seeds), against single-source transfer and the in-domain ceiling.

Held out	mean single	best single	pooled	+ masking	INLP	CORAL	in-domain
ISOT	0.670	0.847	0.865	0.890	0.805	0.781	0.999
WELFake	0.540	0.690	0.648	0.657	0.643	0.643	0.989
PolitiFact	0.592	0.730	0.677	0.655	0.600	0.650	0.893
GossipCop	0.522	0.544	0.538	0.546	0.558	0.520	0.869
LIAR	0.530	0.552	0.528	0.528	0.502	0.514	0.698
Twitter15	0.563	0.657	0.647	0.645	0.592	0.626	0.981
Twitter16	0.552	0.678	0.715	0.703	0.536	0.616	0.967

The ranking part is hard. Source diversity helped: pooled training reached 0.660 held-out AUC against 0.567 for a single source. But label-free invariance methods made ranking worse, and the pooled model is still far below the in-domain ceiling of 0.914. Invariance to corpus identity is not enough when the label itself is tied to how each corpus was put together. Better ranking will need better data, in particular corpora whose real and fake items come from the same outlets and time windows, or evidence-based detectors that do not lean on style (Pelrine et al., 2023; Thibault et al., 2025). Multi-domain architectures (Nan et al., 2021; Zhu et al., 2022) should be evaluated on held-out corpora with this entanglement in mind.

Style as construct and as shortcut. The Bengali results add a distinction that the English matrix alone would not show. Clickbait detection held up on unseen publishers and partly across a change of language, while veracity detection did not survive a change of corpus. A plausible explanation is that a style-sensitive model transfers when style *is* the construct, as with clickbait headlines, and fails when style is only correlated with

Table 12: BanglaBait, official split. Clickbait-class F1, precision and recall and accuracy (%), as reported by Mahtab et al. (2023), plus ROC-AUC. Reference values from that paper (as released): LR char 3–5-grams F1 66.28, BanglaBERT 71.72, SS-GAN BanglaBERT 75.13.

Test set	Input	Model	F1	P	R	Acc.	AUC
as released	title	LR, char 3–5	64.37	73.63	57.17	77.72	0.834
as released	title	SVM, char 3–5	63.79	73.37	56.42	77.46	0.836
as released	title	LR, word 1–2	48.51	65.59	38.49	71.25	0.737
as released	title+content	LR, char 3–5	61.29	71.92	53.40	76.26	0.792
as released	title+content	SVM, char 3–5	60.68	71.78	52.55	76.03	0.791
as released	title+content	LR, word 1–2	57.48	70.36	48.58	74.70	0.775
deduplicated	title	LR, char 3–5	59.15	70.72	50.84	76.57	0.823
deduplicated	title	SVM, char 3–5	60.01	71.23	51.85	76.94	0.824
deduplicated	title	LR, word 1–2	40.77	64.56	29.79	71.11	0.722
deduplicated	title+content	LR, char 3–5	52.34	67.98	42.55	74.14	0.772
deduplicated	title+content	SVM, char 3–5	53.28	69.42	43.23	74.70	0.772
deduplicated	title+content	LR, word 1–2	48.61	66.86	38.19	73.06	0.752

the label inside one collection, as with fake news. With a single Bengali corpus and labels made in different ways, we offer this as a hypothesis for benchmark designers, not a settled result: they should say which kind of cue their labels license.

Reporting protocol. For any claim that a misinformation detector generalises: (1) *verify label direction* with signature probes, and treat a same-task AUC below 0.5 as a data error until shown otherwise; (2) *deduplicate* within and across corpora with near-duplicate matching, and drop label-conflicting groups; (3) report at least one *same-task cross-corpus* result, with a ranking metric (ROC-AUC or AP) next to F1; (4) report the *decomposition*: F1 at the default, prior-matched and (cross-fitted) oracle thresholds; (5) run three cheap *shortcut probes*: a surface-statistics model, a group-identity baseline where metadata exist, and a 64-example learning-curve point; (6) report a *permutation control*, several seeds, and intervals that resample corpora when averaging over a matrix; (7) deploy with a *rate-based* operating point unless target labels are available, and estimate the target base rate from a labelled sample if it is expected to change.

6.1. Limitations and threats to validity

Models. Six of our detectors are linear or tree-based on sparse, static-embedding and frozen RoBERTa features. The seventh fine-tunes only the upper four layers of RoBERTa-base, on 800 training items per corpus and three seeds, because the study ran on a two-core CPU; full fine-tuning on the complete splits might do better in-domain, though prior work found that it does not reliably help across corpora (Verma,

2026; Dell'Oglio et al., 2026). We did not prompt large language models, which earlier studies found more robust across corpora (Pelrine et al., 2023; Dell'Oglio et al., 2026). The decomposition applies unchanged to any detector that outputs a score, and testing it on those families is the most direct extension. *Corpora.* Labels differ in kind (claims, stories, outlets), which is why we lean on same-task pairs. FakeNewsNet is used through its public titles only. NELA-GT and BanFakeNews could not be obtained in our environment, so the Bengali analysis uses one Bengali corpus and a translate-test rather than a second native corpus. PHEME labels rumours, not falsehood, and serves only as an additional unit. The article corpora enter as cleaned working copies (Section 3). Deduplication thresholds (Jaccard ≥ 0.8) trade recall for precision, and looser thresholds would remove more items. *Statistics.* Twitter15/16 and PolitiFact test sets are small, so their cells have wide intervals (Table 5). We report pooled seed-bootstrap intervals per cell, resample corpora for matrix-level quantities, and avoid conclusions from single small cells. Seven corpora are still few, and the corpus-level intervals are correspondingly wide. *Operating points.* Rate transfer worked here partly because the base rates of our corpora are similar. The prior-shift experiment shows how it degrades when they are not; that experiment resamples existing test sets and does not reproduce every way a real feed can change. *Scope.* Our results concern text-only detection. Propagation-based detectors face related spurious-correlation problems on Twitter15/16 and PHEME (Wu and Hooi, 2022; Mu et al., 2023; Pelrine et al., 2021), which we did not re-evaluate. *Ethics.* We use only released text of public posts, with no user identifiers and no re-collection. Tweet content remains subject to the platform's terms, so we release code, per-cell results and saved scores but no post text.

7. Conclusion

Misinformation detectors that look near-perfect in-domain lose much of their value on a new corpus, though not all of it, and not for a single reason. Across ten audited corpora and seven detectors, 70% of the macro-F1 they lose is a ranking deficit that only better data or representations can address. The remaining 30% is a misplaced operating point, and most of it can be repaired without a single target label by transferring the share of items the detector flags rather than trusting its scores or a label-shift correction. That repair is exact when target scores are an order-preserving distortion of source scores and the base rate is unchanged, and it weakens as the base rate moves. Much of what in-domain benchmarks reward is the collection process behind each class: outlet signatures, attribution style, time windows. Where style is the construct itself, as with clickbait, detection travels across publishers and in part across languages; where it is a by-product of how a corpus was assembled, as with veracity, it does not. Papers that claim generalisation should report transfer, ranking metrics and the decomposition next to in-domain accuracy, and the protocol and code released with this paper make that cheap to do.

CRediT authorship contribution statement

Dewan Fahad Chowdhury: Conceptualization, Methodology, Software, Formal analysis, Investigation, Data curation, Writing – original draft, Visualization. **Md. Rejaul Korim Sadi:** Conceptualization, Methodology, Validation, Investigation, Data curation, Writing – review & editing, Project administration.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data and code availability

All corpora are publicly available from their original releases (Section 3 and the repository's `DATA.md`); none is redistributed. Code, fixed seeds, per-cell results, saved predictions, logs and a one-command reproduction script are released under the MIT licence at `https://github.com/<account>/fn-transfer-audit` (archived release DOI to be added on acceptance).

Declaration of generative AI and AI-assisted technologies in the manuscript preparation process

During the preparation of this work the authors used Claude (Anthropic) to assist with code development, literature search and drafting of the text. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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Appendix A. Proof of Proposition 1

Write $M^A(u) = \pi_A F_1^A(u) + (1-\pi_A)F_0^A(u)$ for the distribution function of source scores and M^B for that of target scores, and assume M^A is continuous and strictly increasing around τ_0, so that quantiles are unique. The source model flags $q_A = 1 - M^A(\tau_0)$ of the source items.

Rate transfer under (i) and (ii). By (i) and (ii), $M^B(g(u)) = \pi_A F_1^B(g(u)) + (1-\pi_A)F_0^B(g(u)) = M^A(u)$ for every u. The threshold that flags the top q_A share of target items is the t with $M^B(t) = 1-q_A = M^A(\tau_0)$, that is, $t = g(\tau_0)$. Its true-positive rate on B is $1 - F_1^B(g(\tau_0)) = 1 - F_1^A(\tau_0)$, the source true-positive rate at τ_0, and the same argument holds for the false-positive rate. With equal priors, equal error rates give the same expected confusion matrix, and hence the same population macro-F1 and balanced accuracy. The fixed threshold τ_0 has true-positive rate $1 - F_1^A(g^{-1}(\tau_0))$ on B, which equals the source rate for every pair of class distributions only if $g^{-1}(\tau_0) = \tau_0$. EM is also misled under (i): it assumes $F_y^B = F_y^A$, so a monotone distortion of the scores is read as a change of class balance.

Pure label shift. If g is the identity and $\pi_B \neq \pi_A$, then $p_B(x \mid y) = p_A(x \mid y)$, calibrated source posteriors are related to target posteriors by Bayes' rule with the two priors, and the EM fixed point of Saerens et al. (2002) is the true π_B in the population limit. The share of target items above τ_0 is $\pi_B \mathrm{TPR}_A + (1-\pi_B)\mathrm{FPR}_A$, which differs from q_A whenever $\mathrm{TPR}_A \neq \mathrm{FPR}_A$. Rate transfer therefore moves the threshold away from τ_0 even though the class-conditional score distributions have not changed, so it no longer reproduces the source error rates, and in general it does not coincide with the Bayes threshold for π_B either. □

Appendix B. Implementation details and compute

Hyperparameters. Logistic regression and linear SVM: $C \in \{0.25, 1, 4\}$ (SVM: $C/4$), liblinear, selected by validation ROC-AUC; TF-IDF word 1–2-grams, `min_df`=2, 100,000 features, sublinear term frequency, texts truncated to 20,000 characters. NB-SVM: binary 1–2-grams, add-one smoothed log-count ratios. GloVe: 100-d `glove-wiki-gigaword`, mean of in-vocabulary tokens. Style model: histogram gradient boosting with 100 or 300 iterations, learning rate 0.1. Frozen RoBERTa: mean of last-layer piece states, standardised, logistic regression with $C \in \{0.025, 0.1, 0.4\}$. Fine-tuned RoBERTa: lower eight layers frozen and cached, upper four layers and a linear head trained with AdamW for three epochs (details in the Supplementary Material). Platt scaling on the validation split for every model. Bengali: character 3–5-grams (`char_wb`), 200,000 features.

Deduplication. MinHash with 128 permutations (seed 1) and LSH at Jaccard 0.8; word 3-gram shingles over the first 400 words for articles, character 5-grams otherwise; text normalised with NFKC, lower-casing, URL and punctuation removal.

Compute. All experiments ran on a 2-core CPU with 8 GB RAM and no GPU. The full pipeline (`run_all.sh`) takes about ten to twelve hours, plus roughly four hours for caching and fine-tuning the transformer. Software: Python 3.13, scikit-learn 1.9.1, spaCy 3.8, PyTorch 2 (CPU), datasketch, statsmodels.

Appendix C. Reproducibility checklist

- Data: public sources, licences and expected layout in `DATA.md`; label harmonisation and its verification in `src/label_probe.py`; deduplication counts in `results/dedup_report.json`.
- Splits and seeds: five seeds (0–4; three for the fine-tuned transformer) fix splits, subsampling and model initialisation; vectorisers and calibrators are fitted on source training and validation data only.
- Every number in this paper is written by `src/paper_numbers.py` from the files in `results/`; no number is typed by hand.
- Saved per-item scores for every (model, source, target, seed) cell allow any metric or threshold rule to be recomputed without retraining.
- Sanity controls: permutation regime (AUC ≈ 0.5) and size-matched regime are part of the main grid.
- The Supplementary Material contains the full matrices for every model, per-cell decompositions and operating points, learning curves, the vocabulary ablation, per-publisher Bengali results and the deduplication report.